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Evaluation of kinetic models for Fischer–Tropsch cobalt catalysts in a plug flow reactor

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ABSTRACT

In order to do a good and realistic design of a Gas-to-Liquid (GTL) plant, it is necessary to have kinetic models that correctly capture major variations of reaction rates and selectivities subject to changes in process parameters. In this study, a review of published kinetic rate models is done and twelve of them are analyzed based on different criteria, such as their behaviour at high conversions, high water partial pressure, sensitivity to added water, selectivity to C_{5+} products, etc. The rate models are implemented in a plug flow reactor model. Both fixed bed and microchannel reactors can quite accurately be represented by such a model. Here, the main purpose is to see the kinetic effects with changing composition along the reactor. In order to predict the product distribution, we have proposed our own chain growth model and fitted parameters to experimental data from Yang et al. (2016). The chain growth model includes the effect of water and predicts the C_{5+} and methane selectivities quite well.

1. Introduction

There are numerous models available predicting reaction rates of the Fischer–Tropsch synthesis (FTS), and there is a diversity of different model structures. The reasons for the diversity may be assigned to different type of catalyst, type of support and promoters, catalyst particle size and pore size, etc. In Table 1 a summary of 30 selected kinetic studies on cobalt catalysts is presented, dating back to 1949 to present, where different reactor types are applied, in addition to temperature, pressure and H_2/CO ranges. Some of the kinetic models give a detailed description of formation rates of individual components, while most models only describe the overall consumption rate of CO given as simple rate expressions. In Table 2 a summary of the latter type is listed. Some of the models (Yang et al., 1979; Pannell et al., 1980; Wang, 1987) are simple power law expressions:

$$-r_{CO} = aP_{H_2}^{b}P_{CO}^{c} \tag{1}$$

In these expressions b is positive and c is negative, suggesting inhibition by adsorbed CO. Anderson and Emmett (1956) found that the equation developed by Brötz (1949) to be inadequate for data over a wide range of inlet H_2/CO ratios and developed an equation by suggesting that the rate is proportional to the desorption of chains. The concentration of growing chains on the catalyst surface was related empirically to $P_{H_2}^2P_{CO}$. Zennaro et al. (2000) fitted their data fairly well by a simple power law expression of the form:

$$-r_{CO} = aP_{H_2}^{0.74}P_{CO}^{-0.24} \tag{2}$$

However, they reported that the data are best fitted by a Langmuir–Hinshelwood (LH) rate model similar to that proposed by Yates and Satterfield (1991).

$$-r_{CO} = \frac{aP_{CO}P_{H_2}^{0.74}}{(1+bP_{CO})^2} \tag{3}$$

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Table 1 – Kinetic studies of Co-based catalysts for Fischer–Tropsch synthesis.

No.	Study	Year	Catalyst	Reactor	T (°C)	P (bar)	H_2/CO
1	Brötz (1949)	1949	$Co/MgO/ThO_2$/kieselguhr	FBR[a]	185–200	1	2
2	Anderson and Emmett (1956)	1956	Co/ThO_2/kieselguhr	FBR	186–207	1	0.9–3.5
3	Yang et al. (1979)	1979	$Co/CuO/Al_2O_3$	FBR	235–270	1.7–55	1.0–3.0
4	Pannell et al. (1980)	1980	$Co/La_2O_3/Al_2O_3$	Berty (low conversion)	215	5.2–8.4	2
5	Outi et al. (1981)	1981	Co/Al_2O_3	FBR (low conversion)	250	1	0.2–4.0
6	Wang (1987)	1987	$Co/B/Al_2O_3$	FBR (low conversion)	170–195	1–2	0.25–4.0
7	Wojciechowski (1988)	1988	Co/kieselguhr	Berty	190	2–15	0.5–8.3
8	Sarup and Wojciechowski (1989)	1989	Co/kieselguhr	Berty	190	2–15	0.5–8.3
9	Withers et al. (1990)	1990	$Co/Zr/SiO_2$	CSTR[b]	240–280	30	1.0–2.0
10	Yates and Satterfield (1991)	1991	$Co/MgO/SiO_2$	Slurry	220–240	5–15	1.5–3.5
11	Iglesia and Sebastian (1993)	1993	Co catalysts	FBR	200–210	1–30	1–10
12	Van Steen and Schulz (1999)	1999	$Co/MgO/ThO_2/SiO_2$	CSTR	190–210	2.7–6.2	Varied widely with water added
13	Zennaro et al. (2000)	2000	Co/TiO_2	FBR	200	8–16	1–4
14	Elbashir and Roberts (2004)	2004	Co/Al_2O_3	FBR – supercritical	230–260	60	0.5–2.0
15	Das et al. (2005)	2005	Co/SiO_2	CSTR	210	21.7	1–2.4
16	Storsæter et al. (2006)	2006	Co/Al_2O_3	Microreactor	210,220	1.85	10
17	Visconti et al. (2007)	2007	Co/Al_2O_3	FBR – micro reactor	210–235	8–25	1.8–2.7
18	Irankhah et al. (2007)	2007	$Co–Ru/Al_2O_3$	FBR	235–250	45–65	1.7–2.3
19	Botes et al. (2009)	2009	$CO/Pt/Al_2O_3$	Slurry	230	5–40	1.6–3.2
20	Atashi et al. (2010)	2010	$Co–Mn/TiO_2$	FBR – micro reactor	190–280	1.0–10	1.0–3.0
21	Bhatelia et al. (2011)	2011	$Ru/Co/Al_2O_3$	CSTR	205–220	14–24	1.4–2.1
22	Visconti et al. (2011)	2011	Co/Al_2O_3	FBR – micro reactor	210–236	8–26	1.8–2.8
23	Mansouri et al. (2013)	2013	$Co–Ce/SiO_2$	FBR – micro reactor	200–300	1	1.0–1.5
24	Nikparsaa et al. (2014)	2014	$Co–Ni/Al_2O_3$	FBR	250	30	2
25	Todic et al. (2014, 2015, 2013)	2014	Co/Re	Slurry	205–210	15–25	1.4–2.1
26	Ma et al. (2014)	2014	Co/Al_2O_3	CSTR	205–230	14–25	1.0–2.5
27	Mosayebi and Haghtalab (2015)	2015	$Co–Ru/\gamma-Al_2O_3$	FBR	220–250	15–25	1.0–2.0
28	Azadi et al. (2015)	2015	$Co/\gamma-Al_2O_3$	Carberry spinning basket batch reactor	196–211	20	1.8–2.9
29	Mosayebi et al. (2016)	2016	$Co–Ru/\gamma-Al_2O_3$	FBR	200–240	10–30	1–3
30	Keyvanloo et al. (2016)	2016	$Co–Si/\gamma-Al_2O_3$	FBR	210–240	20	1.3–3.0

[a] FBR: fixed bed reactor.
[b] CSTR: continuous stirred tank reactor.

Yates and Satterfield (1991) proposed the following rate model:

$$-r_{CO} = \frac{aP_{CO}P_{H_2}}{(1 + bP_{CO})^2} \tag{4}$$

They correctly pointed out that it is often a high degree of covariance between the estimated parameters, so that several terms in the denominator can barely be justified from a statistical perspective. In terms of Langmuir–Hinshelwood (LH) kinetics, one of these parameters would represent a surface rate constant while others are adsorption equilibrium coefficient. It is assumed that CO is the predominant surface species, which is justified by non-reacting, single component adsorption data on cobalt surfaces.

Botes et al. (2009), in their search for a suitable model, tested 16 different reaction models and came up with an LH kinetic model. Their preferred kinetic model is considered semi-empirical because it cannot be derived from an assumed reaction mechanism.

Keyvanloo et al. (2016) explored several mechanisms and LH rate models. Their proposed rate is:

$$-r_{CO} = \frac{aP_{CO}^{0.5}P_{H_2}^{1}}{(1 + bP_{CO} + cP_{H_2}^{0.5})^2} \tag{5}$$

In the denominator b is an activated rate constant and not an equilibrium constant. Therefore b increases with increasing temperature instead of decreasing. Ma et al. (2014) observed a positive kinetic water effect on Co/Al_2O_3 catalyst. It is consistent with the results of the effect of co-fed water on cobalt FTS in this work and the literature (e.g. Lögdberg et al. (2011)) for Co/Al_2O_3 catalysts. Ma et al. (2014) employed their empirical model which has the effects of water:

$$-r_{CO} = \frac{k_1(T)P_{CO}^{-0.31}P_{H_2}^{0.88}}{1 - 0.24\frac{P_{H_2O}}{P_{H_2}}} \tag{6}$$

Some researchers have proposed more detailed kinetic models which are more complex and include a description of product distribution. The works of Storsæter et al. (2006), Visconti et al. (2007, 2011), Todic et al. (2013, 2014, 2015), Mosayebi and Haghtalab (2015) and Mosayebi et al. (2016) are among this category.

The role of water on catalytic activity is a highly complex matter. Water is known to increase or decrease the catalytic activity. The influence of water on the cobalt-FT reaction kinetics has become an important topic in the literature and some authors have started to include water in their kinetic models (Withers et al., 1990; Van Steen and Schulz, 1999; Das et al., 2005; Bhatelia et al., 2011; Ma et al., 2014). The results

Table 2 – Simple rate models proposed for cobalt FTS.

No.	Study	Rate model
1	Brötz (1949)	$-r_{CO} = \frac{aP_{H_2}^2}{P_{CO}}$
2	Anderson and Emmett (1956)	$-r_{CO} = \frac{aP_{H_2}^2 P_{CO}}{1+bP_{H_2}^2 P_{CO}}$
3	Yang et al. (1979)	$-r_{CO} = aP_{H_2} P_{CO}^{-0.5}$
4	Pannell et al. (1980)	$-r_{CO} = aP_{H_2}^{0.55} P_{CO}^{-0.33}$
5	Outi et al. (1981)	$-r_{CO} = \frac{aP_{H_2}^1 P_{CO}^{0.5}}{(1+bP_{CO}^{0.5})^3}$
6	Wang (1987)	$-r_{CO} = aP_{H_2}^{0.66} P_{CO}^{-0.5}$
7	Wojciechowski (1988)	$-r_{CO} = \frac{aP_{H_2}^{0.5} P_{CO}^1}{(1+bP_{CO}^1+cP_{H_2}^{0.5})^2}$
8	Sarup and Wojciechowski (1989)	$-r_{CO} = \frac{aP_{H_2}^{0.5} P_{CO}^{0.5}}{(1+bP_{CO}^{0.5}+cP_{H_2}^{0.5}+dP_{CO})^2}$
9	Withers et al. (1990)	$-r_{CO} = \frac{aP_{H_2}}{1+b\frac{P_{H_2O}}{P_{CO}P_{H_2}}}$
10	Yates and Satterfield (1991)	$-r_{CO} = \frac{aP_{H_2}^1 P_{CO}^1}{(1+bP_{CO}^1)^2}$
11	Iglesia and Sebastian (1993)	$-r_{CO} = \frac{aP_{H_2}^{0.6} P_{CO}^{0.65}}{1+bP_{CO}^1}$
12	Van Steen and Schulz (1999)	$-r_{CO} = \frac{kP_{H_2}^{1.5}\frac{P_{CO}}{P_{H_2O}}}{(1+kP_{H_2}(\frac{P_{CO}}{P_{H_2O}}))^2}$
13	Zennaro et al. (2000)	$-r_{CO} = \frac{aP_{H_2}^{0.74} P_{CO}^1}{(1+bP_{CO}^1)^2}$
13	Zennaro et al. (2000)	$-r_{CO} = aP_{H_2}^{0.74} P_{CO}^{-0.24}$
14	Elbashir and Roberts (2004)	$-r_{CO} = \frac{aP_{H_2}^{0.5} P_{CO}^{0.5}}{(1+bP_{CO}^{0.5}+cP_{H_2}^{0.5}+dP_{CO})^2}$
15	Das et al. (2005)	$-r_{CO} = \frac{aP_{CO}^{-0.25} P_{H_2}^{0.5}}{1+m\frac{P_{H_2O}}{P_{H_2}}}$
18	Irankhah et al. (2007)	$-r_{CO} = \frac{aP_{H_2}^{0.5} P_{CO}^{0.5}}{(1+bP_{CO}^{0.5}+cP_{H_2}^{0.5}+dP_{CO})^2}$
19	Botes et al. (2009)	$-r_{CO} = \frac{aP_{CO}^{0.5} P_{H_2}^{0.75}}{(1+bP_{CO}^{0.5})^2}$
20	Atashi et al. (2010)	$-r_{CO} = \frac{aP_{CO}P_{H_2}}{1+bP_{CO}}$
20	Atashi et al. (2010)	$-r_{CO} = \frac{aP_{CO}P_{H_2}^2}{(1+2(bP_{CO})^{0.5})^2}$
21	Bhatelia et al. (2011)	$-r_{CO} = \frac{aP_{H_2}^{0.5} P_{CO}}{(1+bP_{CO}+cP_{H_2}^{0.5}+mP_{H_2O})^2}$
23	Mansouri et al. (2013)	$-r_{CO} = \frac{aP_{H_2}}{(1+bP_{CO})^2}$
24	Nikparsaa et al. (2014)	$-r_{CO} = \frac{aP_{H_2}P_{CO}}{(1+bP_{CO}P_{H_2}^{0.5})^2}$
26	Ma et al. (2014)	$-r_{CO} = \frac{aP_{CO}^{-0.31} P_{H_2}^{0.88}}{1+m\frac{P_{H_2O}}{P_{H_2}}}$
30	Keyvanloo et al. (2016)	$-r_{CO} = \frac{aP_{CO}^{0.5} P_{H_2}}{(1+bP_{CO}+cP_{H_2}^{0.5})^2}$

of water co-feeding studies have not been very consistent, as some reported a positive effect on the reaction rate (Schulz et al., 1994; Krishnamoorthy et al., 2002; Li et al., 2002) and some a negative influence of water on the reaction rate (Li et al., 2002), while others have found no effect at all (Schulz et al., 1997; Li et al., 2002). According to Ma et al. (2014), a positive water effect on the FT rate was observed on SiO_2 supported Co catalysts (Krishnamoorthy et al., 2002; Li et al., 2002; Storsæter et al., 2005), a positive (Storsæter et al., 2005) or a slightly negative (Li et al., 2002) water effect was observed on Co/TiO_2 catalysts, and a negative (Li et al., 2002) or no significant effect (Schulz et al., 1997; Botes, 2009) or a positive water effect (Borg et al., 2006; Lögdberg et al., 2011) was observed on Co/Al_2O_3 catalysts. Dalai and Davis (2008) reviewed water effects on the performances of unsupported and supported Co catalysts. They concluded that the effects of water on FTS is quite complex and depends on the support and its nature, Co metal loading, its promotion with noble metals, and preparation procedure.

The product distribution from the Fischer–Tropsch synthesis is often described by the Anderson–Schulz–Flory (ASF) distribution, where the distribution is given by a single parameter; the chain growth probability or growth factor (α). In order to perform an FT reactor design with realistic evaluation of the products produced, there is a need for a way to quantify the product distribution. Some kinetic models include a description of product distribution. However most of them need a chain growth model to account for product distribution. Hydrocarbon selectivities depend on local conditions like temperature, pressure and species concentrations, in particular of carbon monoxide and hydrogen, but also on produced water. Therefore a suitable chain growth model should include these variables. The effect of some of the most important parameters affecting product selectivity are discussed thoroughly in literature by for example Van Der Laan and Beenackers (1999).

According to Ma et al. (2014), the effect of water on selectivities to C_{5+} and CH_4 for cobalt based catalyst has reached a consensus; indigenous or added water increase C_{5+} selectivity and decrease CH_4 selectivity (Schulz et al., 1997; Krishnamoorthy et al., 2002; Storsæter et al., 2005; Botes, 2009; Ma et al., 2011). Lögdberg et al. (2010) investigated the effect of adding water vapor to 20 cobalt-based supported catalysts. Although there was a large variation in selectivity, added water increased C_{5+} selectivity and decreased CH_4 selectivity for all catalysts. It is surprising however, that none of the available chain growth models in literature have water partial pressure as a variable. In this article we propose a new chain growth model which includes the effect of water and predicts the C_{5+} and methane selectivities quite well.

2. Chain growth models

2.1. *From literature*

Only few detailed product distribution models have been proposed for Co-based FTS (Visconti et al., 2007, 2011; Kwack et al., 2011; Todic et al., 2013, 2014, 2015; Mosayebi and Haghtalab, 2015; Mosayebi et al., 2016). Many of the chain growth models available are parts of micro kinetic models where chain growth probability changes with the propagation step. The growth factor is the ratio between two consecutive product formation rates which is a constant. However, what is often observed is a distribution that can be described by two or more growth factors. A possible model is to let the growth factor α_1 be the larger to produce paraffins, and the growth factor α_2, be the smaller to produce olefins. The FT reactions may be lumped into one reaction, and if all products with five and more carbons is lumped into a single C_{5+} component, the reaction can be written as:

$$CO + U \cdot H_2 \xrightarrow{r_{FT}} H_2O + \nu_1 C_1 + \nu_2 C_2 + \nu_3 C_3 + \nu_4 C_4 + \nu_{5+} C_{5+} \quad (7)$$

In addition, a known deviation from ideal ASF distribution is the enhanced production of methane, which can be accounted for by introducing an extra methanation reaction. How to calculate the stoichiometric coefficients U, ν_i and average carbon number of the lump is shown by Hillestad (2015). The stoichiometric product coefficients are $\nu_i = (1-\alpha)^2\alpha^{i-1}$ and $\nu_{5+} = (1-\alpha)\alpha^4$. The stoichiometric usage ratios of hydrogen, U, for production of paraffins and olefins are:

$$U_1 = 3 - \alpha_1 \tag{8}$$

$$U_2 = 2 + (1-\alpha_2)^2 \tag{9}$$

Depending on α_1 and α_2, the stoichiometric usage ratio will be slightly above 2. If the amount of H_2 in the feed is more than the stoichiometric usage ratio, CO will be the limiting reactant and the H_2/CO ratio will increase along the reactor. On the other hand, if the amount of H_2 is below the stoichiometric usage ratio, H_2 will be the limiting reactant and the H_2/CO ratio will decrease along the reactor.

A very simple chain growth model is suggested by Yermakova and Anikeev (2000) as an empirical relation between α and the compositions of H_2 and CO after testing several different equation forms:

$$\alpha = A\frac{y_{CO}}{y_{H_2} + y_{CO}} + B \tag{10}$$

Here, y_{CO} and y_{H_2} are the mol fractions of CO and H_2 in the gas phase. They fitted this model to their experimental data over an alumina-supported cobalt catalyst promoted with zirconium, and estimated constants A and B to be 0.2332 ± 0.0740 and 0.6330 ± 0.0420, respectively. This model lacks temperature dependency and Song et al. (2004) modified the model to:

$$\alpha = \left(A\frac{y_{CO}}{y_{H_2} + y_{CO}} + B\right)[1 - 0.0039(T - 533)] \tag{11}$$

They found the slope of temperature dependency by fitting Eq. (11) to various experimental data in the review paper by Van Der Laan and Beenackers (1999). However, there is a concern over this temperature correlation in that the data in the mentioned reference are related to Fe and Ru catalysts which are different from the original catalyst for which Eq. (10) is developed. Lox and Froment (1993) proposed a chain growth model for an iron catalyst which is a function of temperature:

$$\alpha = \frac{k_1(T)P_{CO}}{k_1(T)P_{CO} + k_2(T)P_{H_2} + k_3(T)} \tag{12}$$

Vervloet et al. (2012) derived a chain growth model for a cobalt catalyst:

$$\alpha = \frac{1}{1 + k_\alpha\left(\frac{C_{H_2}}{C_{CO}}\right)^{\beta}\exp\left(\frac{\Delta E}{R}\left(\frac{1}{493.15} - \frac{1}{T}\right)\right)} \tag{13}$$

The model does not take into account the effect of water. Kruit et al. (2013) and Becker et al. (2016) have used this model in their work to explain the deviations from the ASF distribution by using an α which is dependent on process conditions.

Ermolaev et al. (2015) used the following growth model in their fixed bed model with cobalt catalysts:

$$\alpha = \frac{1}{1 + \frac{k_\alpha}{p_{CO}^{2/3}p_{H_2}^{2/3}}\cdot\left(1 + \frac{1}{1 + k_\beta p_{H_2}^{0.5}}\right)} \tag{14}$$

where

$$k_\alpha = 40.3\exp\left(-9600\left(\frac{1}{T} - \frac{1}{488.15}\right)\right) \tag{15}$$

$$k_\beta = 0.035\exp\left(-4800\left(\frac{1}{T} - \frac{1}{488.15}\right)\right) \tag{16}$$

All parameters, such as pre-exponential factors and activation energies, were obtained from experimental data of laboratory scale fixed bed reactor.

Extensions to the ASF model has been made by Förtsch et al. (2015) to account for deviations from ASF distribution. These extensions includes a description of enhanced product formation and re-adsorption. However, estimating several parameters from experimental data can be uncertain especially in case of deviations from ideal experimental conditions, such as diffusion limitations, existence of local hot spots or product accumulation in the reactor (Förtsch et al., 2015).

2.2. *Our proposed model*

In order to predict the product distribution, we have developed a chain growth model based on the experimental data of Yang et al. (2016). They performed a kinetic investigation on 40%Co/1%Re/γ-Al_2O_3 catalyst at reactor conditions of 210–240 °C, 20–40 bar and feed H_2/CO = 2.17. They studied the effect of operation temperature, pressure and gas-hourly space velocity on CO conversion, C_{5+} selectivity, CH_4 selectivity and stability during Fischer–Tropsch synthesis in a microchannel reactor. The model structure is based on the propagation probability being equal to the ratio of propagation rate and the sum of propagation and termination rates. Empirical exponents are introduced and only one temperature parameter is estimated. The fitted model is:

$$\alpha = \frac{1}{1 + k_2(T)\frac{P_{H_2}^{1.45}}{P_{CO}P_{H_2O}^{0.253}}} \tag{17}$$

$$k_2(T) = 0.0233\exp\left(-1959\left(\frac{1}{T} - \frac{1}{483}\right)\right) \tag{18}$$

The effects of temperature, partial pressures of H_2O, H_2 and CO are taken into account in this model. The growth probability decreases slightly with increasing total pressure, and increases with increasing H_2O partial pressure. With increase in H_2/CO ratio, the growth factor decreases.

A plug flow model is applied to simulate the 18 last experimental points from Yang et al. (2016). These points are from the same catalyst batch. Experimental CO conversion, C_{5+} and CH_4 selectivities range from 20–80%, 76–88% and 6–12%, respectively (see Fig. 1). The kinetic rate model suggested by Ma et al. (2014) is implemented and only one parameter of this rate model is adjusted, the level of k and not the activation energy, as the cobalt loading is different. The rate constant is given by:

$$k(T) = 0.0234\exp\left(-\frac{104,000}{R}\left(\frac{1}{T} - \frac{1}{483}\right)\right) \tag{19}$$

A separate methanation reaction is added to account for the higher methane selectivity than predicted by the ASF distribution. The methanation reaction is found to be

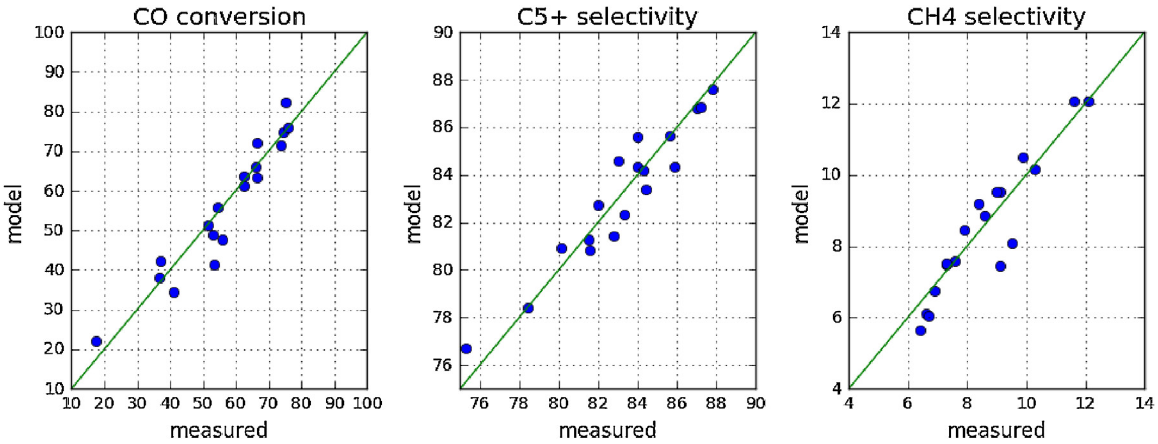

Fig. 1 – Parity plots of measured and estimated CO conversion, C_{5+} and CH_4 selectivities for the developed chain growth model.

proportional to the methane formation rate from reaction (7) and is estimated to be:

$$r_{CH_4} = 7.70 r_{FT} (1-\alpha)^2 \cdot \exp\left(-\frac{36,687}{R}\left(\frac{1}{T}-\frac{1}{483}\right)\right) \tag{20}$$

The olefin production rate is set to 8% of paraffin production rate. To account for the production of olefins, the chain growth factor proposed by Todic et al. (2014) is used:

$$\alpha_2 = \alpha_1 \exp(-0.27) \tag{21}$$

C_1 to C_4 carbon selectivities (S_{C_i}) are calculated as:

$$S_{C_i} = i \times \frac{F_{C_i}^{out} - F_{C_i}^{in}}{F_{CO}^{in} - F_{CO}^{out}} \times 100, \quad i = 1:4 \tag{22}$$

where F_{C_i} is the molar flow of hydrocarbons containing i number of carbon atoms. C_{5+} selectivity is calculated as:

$$S_{C_{5+}} = 100 - \sum_{i=1}^{4} S_{C_i} \tag{23}$$

As a result, the reactor model predicts CO conversion, C_{5+} and CH_4 selectivities quite well, as shown in Figs. 1 and 2.

3. Evaluation of kinetic models

In order to compare different kinetic models, they were implemented in a plug flow model. Our proposed chain growth model (Eq. (17)) is used to calculate selectivities toward C_{5+} and CH_4 as described in Section 2.2. Olefin chain growth factor is calculated with Eq. (21). The olefin production rate is set to 8% of paraffin production rate. To account for higher production of methane than predicted by ASF distribution, Eq. (20) is also implemented. A total of 12 kinetic models mentioned in the work of Ma et al. (2014) is the basis for the present work. They fitted those models to their experimental data on a 25%Co/Al_2O_3 catalyst and calculated parameter values for different kinetic rate models. Here we use the same parameter values so that the different rate models, presented in Table 3, are compared on the same basis.

For each reaction rate model, six different scenarios are investigated which are one over stoichiometric and two under stoichiometric H_2/CO ratios at the inlet of the reactor, each with and without added water. They are shown schematically in Fig. 3. The trends observed for understoichiometric H_2/CO ratios of 2.0 and 1.5 are similar and therefore only results for one of them, namely H_2/CO = 2.0 are shown in the following figures. The reactor is operating at 27 bar and nearly isothermal conditions at 220 °C. Behaviours of rate models and chain growth model as function of H_2/CO ratio, H_2O/H_2 ratio and CO conversion are investigated along the reactor. The effect of CO conversion on C_{5+} selectivity is also studied.

3.1. *Water addition*

In practice, even after the water is knocked out before the FT reactor, the syngas will always contain some water. There has been numerous studies looking into the effects of added water on FTS. For these reasons, it is worthwhile to investigate how different rate models and our chain growth model (Eq. (17)) react to added water in the feed. 20% water is added for this

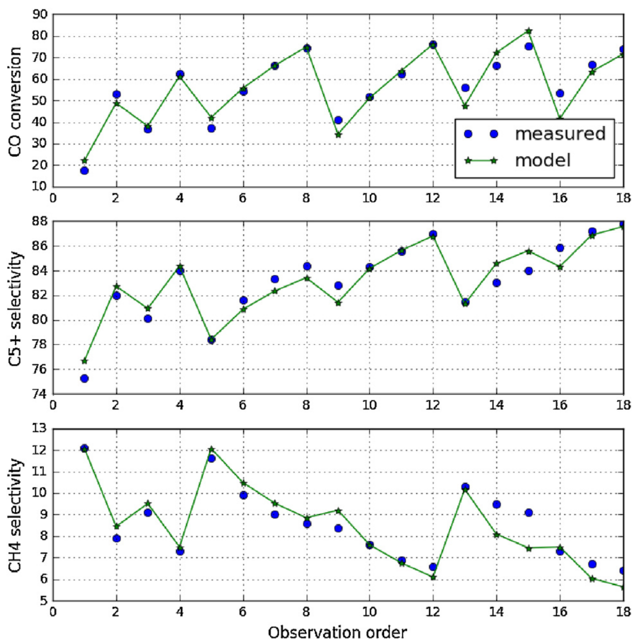

Fig. 2 – Order plot of measured and estimated CO conversion, C_{5+} and CH_4 selectivities for the developed chain growth model.

Table 3 – Kinetic models used (parameter values from Ma et al. (2014)).

Number[a]	Model	Study number in Table 1	Parameter values			
r_1	$-r_{CO} = kP_{CO}^{a}P_{H_2}^{b}$	3, 4, 6, 13	k, mol/g-cat/h/MPa^ (a+b) 0.0137	a −0.35	b 0.81	
r_2	$-r_{CO} = \frac{kP_{CO}^{0.65}P_{H_2}^{0.6}}{(1+K_1P_{CO})}$	11	k, mol/g-cat/h/MPa^ (1.25) 0.6500	K1, MPa^ (−1) 41.30		
r_3	$-r_{CO} = \frac{kP_{CO}P_{H_2}^{0.74}}{(1+K_1P_{CO})^2}$	13	k, mol/g-cat/h/MPa 0.3080	K1, MPa^ (−1) 3.90		
r_4	$-r_{CO} = \frac{kP_{CO}^{0.5}P_{H_2}^{0.5}}{(1+K_1P_{CO}^{0.5}+K_2P_{H_2}^{0.5})^2}$	7, 8	k, mol/g-cat/h/MPa 0.0976	K1, MPa^ (−0.5) 2.36	K2, MPa^ (−1) −0.67	
r_5	$-r_{CO} = \frac{kP_{CO}P_{H_2}^{0.5}}{(1+K_1P_{CO}+K_2P_{H_2}^{0.5})^2}$	7, 8	k, mol/g-cat/h/MPa^ (1.5) 0.0782	K1, MPa^ (−1) 1.95	K2, MPa^ (−1) −0.48	
r_6	$-r_{CO} = \frac{kP_{CO}P_{H_2}}{(1+K_1P_{CO})^2}$	10	k, mol/g-cat/h/MPa^ (2) 0.2696	K1, MPa^ (−1) 3.61		
r_7	$-r_{CO} = \frac{kP_{CO}P_{H_2}^{2}}{(1+K_1P_{CO}P_{H_2}^{2})}$	2	k, mol/g-cat/h/MPa^ (3) 0.1130	K1, MPa^ (−3) 4.83		
r_8	$-r_{CO} = \frac{kP_{CO}^{0.5}P_{H_2}}{(1+K_1P_{CO}^{0.5})^3}$	5	k, mol/g-cat/h/MPa^ (1.5) 24.60	K1, MPa^ (−0.5) 1.65		
r_9	$-r_{CO} = \frac{k(P_{CO}P_{H_2}^{1.5}/P_{H_2O})}{(1+K_1P_{CO}P_{H_2}/P_{H_2O})^2}$	12	k, mol/g-cat/h/MPa^ (2) 0.01470	K1, MPa^ (−1) 0.19		
r_{10}	$-r_{CO} = \frac{kP_{CO}P_{H_2}^{0.5}}{(1+K_1P_{CO}+K_2P_{H_2}^{0.5}+mP_{H_2O})^2}$	21	k, mol/g-cat/h/MPa^ (1.5) 0.0560	K1, MPa^ (−1) 1.61	K12, MPa^ (−0.5) −0.51	m, MPa^ (−1) −0.17
r_{11}	$-r_{CO} = \frac{kP_{CO}^{0.5}P_{H_2}^{0.75}}{(1+K_1P_{CO}^{0.5})}$	19	k, mol/g-cat/h/MPa^ (1.25) 1.36	K1, MPa^ (−0.5) 8.16		
r_{12}	$-r_{CO} = \frac{kP_{CO}^{-0.31}P_{H_2}^{0.88}}{1+m\frac{P_{H_2O}}{P_{H_2}}}$	26	k, mol/g-cat/h/MPa^ (0.57) 0.0133	m −0.24		

[a] Numbers in the following figures refer to these rates.

Table 4 – Inlet mol fractions for different scenarios.

Components	Without water present			With water present		
	Over stoichiometric ($H_2/CO = 2.5$)	Under stoichiometric ($H_2/CO = 2.0$)	Under stoichiometric ($H_2/CO = 1.5$)	Over stoichiometric ($H_2/CO = 2.5$)	Under stoichiometric ($H_2/CO = 2.0$)	Under stoichiometric ($H_2/CO = 1.5$)
H_2	0.714	0.667	0.6	0.5714	0.53333	0.48
CO	0.286	0.333	0.4	0.2286	0.26666	0.32
H_2O	1 ppm[a]	1 ppm[a]	1 ppm[a]	0. 2	0.2	0.2

[a] Very low amount of water to avoid mathematical error in α model and some rates.

investigation. Upon water addition, the total pressure and the flow rate of synthesis gas are kept constant. Thus, the partial pressures of the reactants are reduced as water is introduced in the feed stream. The water flow rate is 20% of the syngas flow rate. Therefore the total flow rate with water addition is 20% higher than the case without water. Inlet compositions for different cases are shown in Table 4.

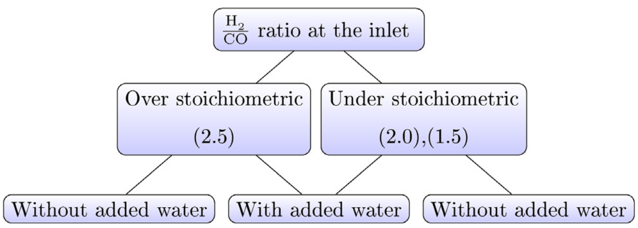

Fig. 3 – Simulation scenarios.

3.2. *Behaviour of chain growth model*

Behaviour of our chain growth model (Eq. (17)) versus CO conversion, H_2/CO ratio and H_2O/H_2 ratio is shown in Fig. 4. The α behaviour as a function of these variables is the same with all kinetic models. It is clearly seen that water has positive effect on α model and increases C_{5+} selectivity. Having an under stoichiometric H_2/CO ratio at the inlet, the ratio decreases along the reactor; while and over stoichiometric H_2/CO ratio at the inlet, gives the opposite effect. Lower H_2/CO ratios yields higher α values. In Fig. 4, it worth to note the behaviour of α model versus CO conversion with overstoichiometric feed. Both curves increase up to CO conversion of about 80% and then they start to decrease. The reasons are attributed to counteracting effects of increasing H_2/CO ratio and water partial pressure on α. In Fig. 4d, the effect of CO conversion on C_{5+} selectivity is shown. It follows the same trend as α model (Fig. 4a).

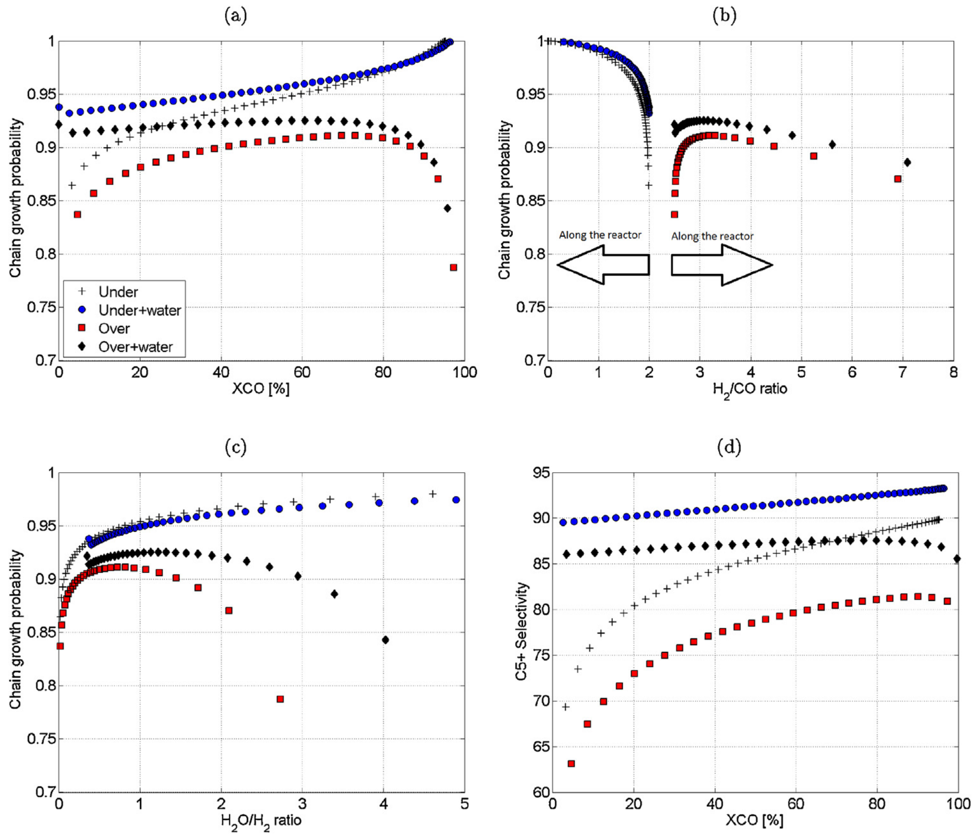

Fig. 4 – (a–c) Chain growth probability as a function of CO conversion, H_2/CO and H_2O/H_2 ratio with under and over stoichiometric H_2 feed; (d) C_{5+} selectivity versus CO conversion with under and over stoichiometric H_2 feed.

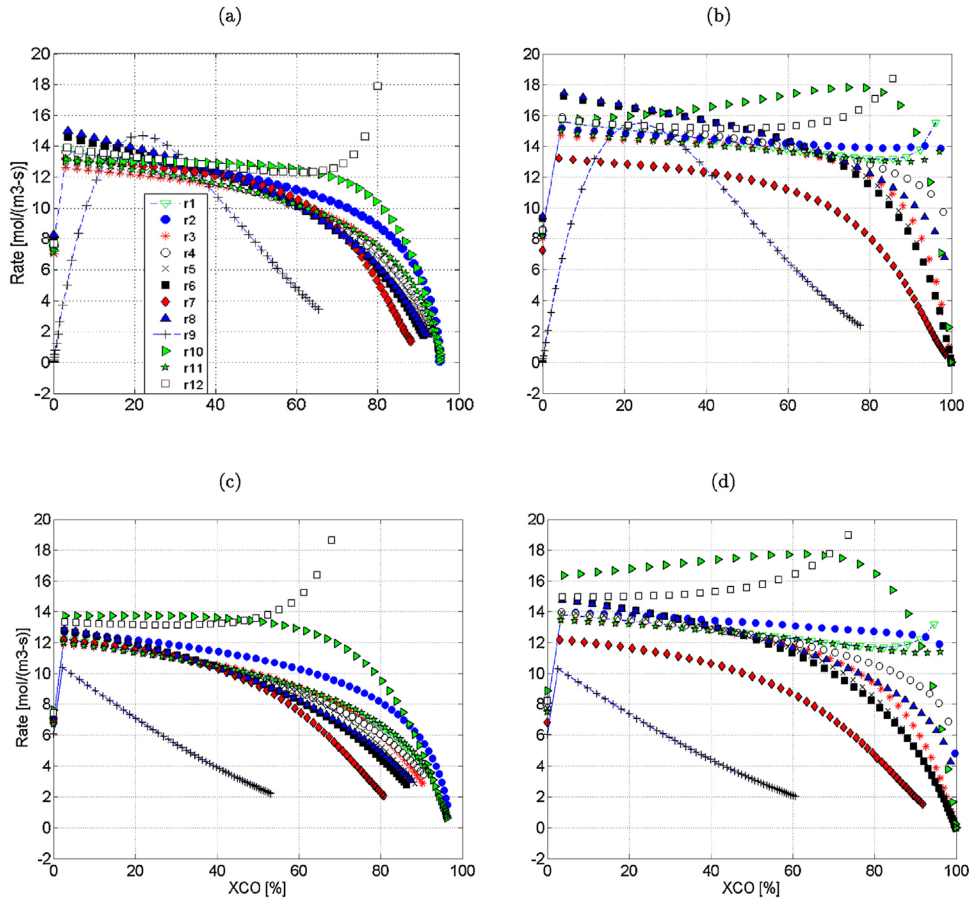

Fig. 5 – Reaction rates versus CO conversion for different rate models with under and over stoichiometric H_2 feed: (a) Under stoichiometric with no added water; (b) Over stoichiometric with no added water; (c) Under stoichiometric with added water; (d) Over stoichiometric with added water.

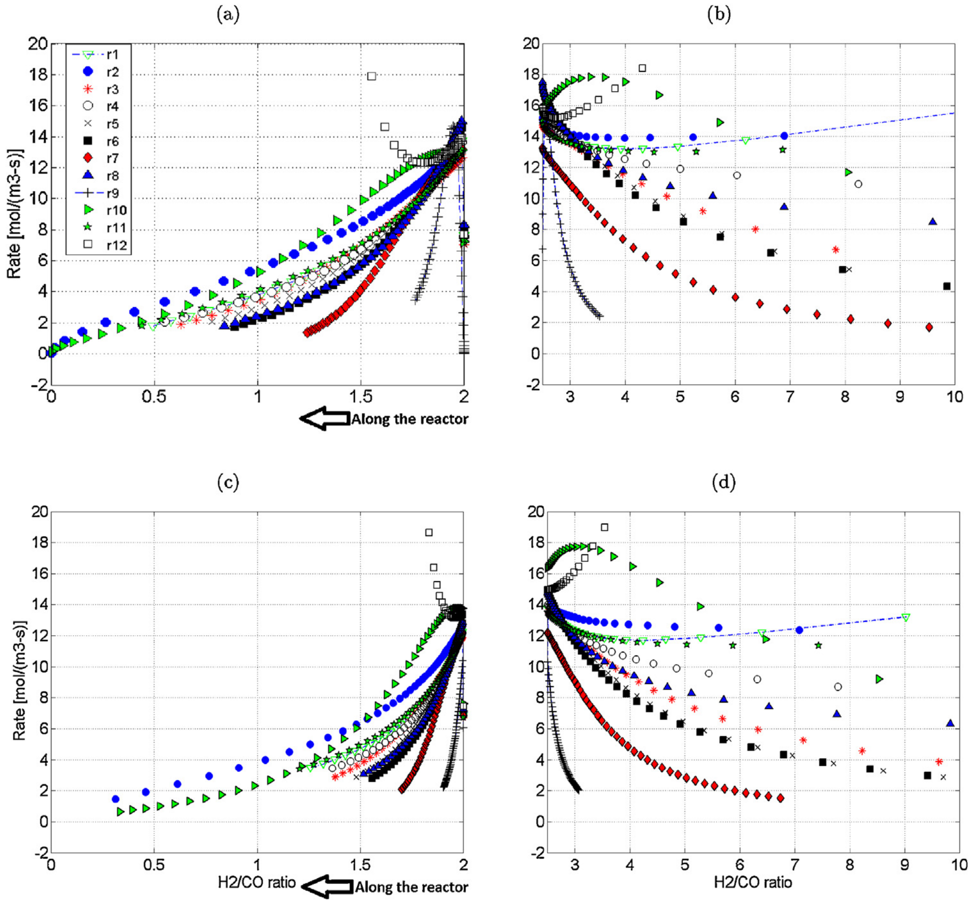

Fig. 6 – Reaction rates versus H_2/CO ratio for different rate models with under and over stoichiometric H_2 feed: (a) Under stoichiometric with no added water; (b) Over stoichiometric with no added water; (c) Under stoichiometric with added water; (d) Over stoichiometric with added water.

3.3. *Behaviour of kinetic rate models*

In Fig. 5, the response of kinetic models as a function of CO conversion along the reactor is plotted. All kinetic models follow the same trend, except r_9, r_{10} and r_{12} which have water term in their structure and with increase of CO conversion, water makes its impacts. r_9 has a bell shape in scenarios with no added water, mainly because of having two water terms in the denominator. r_{10} shows typical behaviour as the rest of the models in under stoichiometric cases, while it increases with conversion before a steep decline as conversion approaches 100% in over stoichiometric cases. r_{12} reaches a singularity point in its denominator at conversions of about 80% which makes it to go to infinity. r_1 goes to infinity at high conversions in over stoichiometric scenarios, because CO which is in the denominator of this rate model gets depleted. One point to mention is that in cases with under stoichiometric H_2/CO ratios at the inlet, none of the rate models reach 100% CO conversion, because H_2 gets depleted and becomes the limiting reactant.

In Fig. 6, different rate models are plotted against the H_2/CO ratio along the reactor. Again all kinetic models follow more or less the same trend except r_9, r_{10} and r_{12}. In under stoichiometric scenarios, H_2/CO ratio decreases along the reactor. In over stoichiometric scenarios, H_2/CO ratio increases along the reactor.

In Fig. 7, different rate models are plotted against the H_2O/H_2 ratio along the reactor. These reflect zones in which catalyst is prone to deactivation with high water partial pressure. Again all kinetic models follow more or less the same trend except r_9, r_{10} and r_{12}.

3.4. *Effect of pressure and added water on FT rate and C_{5+} selectivity at constant conversion (X_{CO} = 50%)*

In order to investigate the effect of pressure on different rates, another pressure level (22 bar) is used and compared with high pressure (27 bar) results. Since the trends for both understoichiometric and over stoichiometric H_2/CO ratio in the feed are similar, only results with understoichiometric feed are presented. The effect of added water on rate models are shown in Fig. 8a. Because of direct effect of pressure, all rates decrease with decreasing pressure. For reaction rates r_1 to r_9, upon water addition, reaction rates decrease. The reason is that by water addition, syngas partial pressure is lowered to keep the total pressure constant. Among all reaction rates, only reaction r_9, r_{10} and r_{12} have water partial pressure present in their structure and water addition shows positive effect on r_{10} and r_{12}.

The effect of pressure and added water on C_{5+} selectivity is shown in Fig. 8b. In our α model (Eq. (17)), C_{5+} selectivities decrease with increasing pressure. The effect of water on C_{5+} selectivities is positive. Added water increased C_{5+} selectivity by as much as 5%. The reason is that our proposed model (Eq. (17)) is strongly dependent on water.

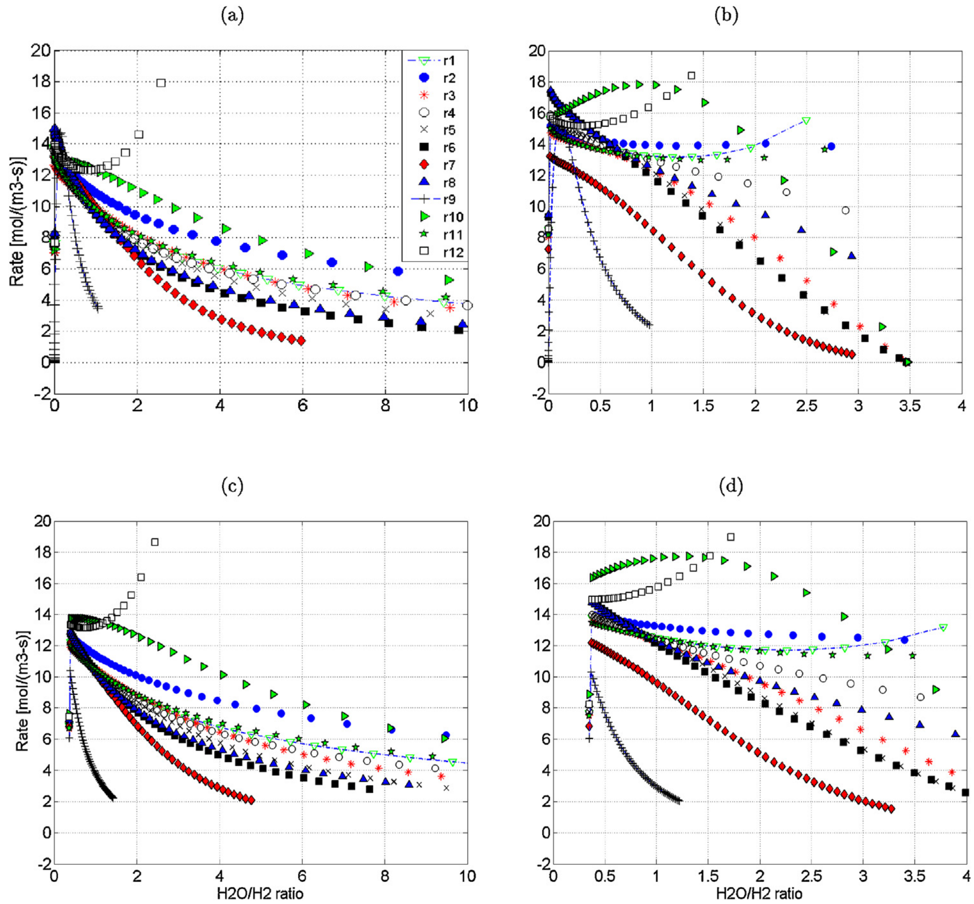

Fig. 7 – Reaction rates versus H_2O/H_2 ratio for different rate models with under and over stoichiometric H_2 feed: (a) Under stoichiometric with no added water; (b) Over stoichiometric with no added water; (c) Under stoichiometric with added water; (d) Over stoichiometric with added water.

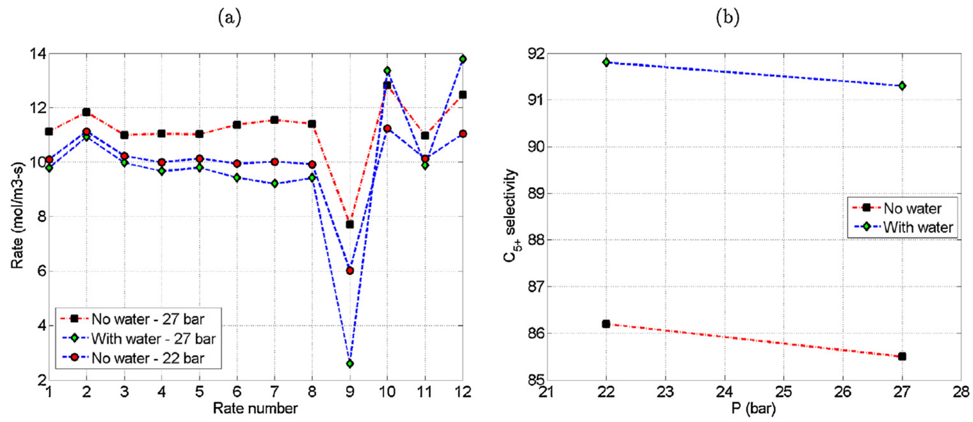

Fig. 8 – Effect of pressure and added water at constant CO conversion of 50% with under stoichiometric H_2/CO ratio feed on: (a) FT rates; (b) C_{5+} selectivity.

4. Conclusions

A new chain growth model is proposed based on microchannel experimental data. This chain growth model together with Ma et al. (2014) kinetic model predicted the experimental data quite well. Twelve kinetic models are implemented in a plug flow reactor model and their behaviour to six different scenarios which are depicted in Fig. 3 are investigated. These kinetic models are compared based on their response to CO conversion, H_2/CO ratio, H_2O/H_2 ratio and C_{5+} selectivities. The trends observed for under stoichiometric ratios of 2.0 and 1.5 are similar. r_9, r_{10} and r_{12} showed different behaviour than the rest of the models, mainly because they have the water effect in their structure. r_{12}, reached a singularity point at CO conversions of about 80%. If it is to be used for reactor design purposes, this should be taken into account. r_1 goes to infinity at very high conversions (more than 95%) in overstoichiometric scenarios, because CO which is in denominator gets depleted. Effects of pressure on rate models and C_{5+} selectivity is also investigated. Two pressure levels (22 and 27 bar) are used for this comparison. Higher pressure yields higher rates for all kinetic models. But the effect of pressure on C_{5+} selectivity is the opposite, higher pressure yields lower C_{5+} selectivity. Effect of added water is also investigated on kinetic models and C_{5+} selectivities. For all rate models except models r_{10} and r_{12}, added water decreased reaction rates, mainly because the syngas pressure is lowered upon water addition to keep the total pressure constant. Upon water addition, C_{5+} selectivities increased by about 5% for all rate models. This study can be an starting point for researchers to look closer to structures of kinetic models.

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